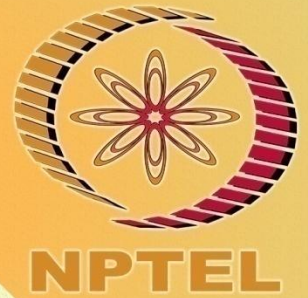


CONCEPTS COVERED

- ☐ PERT
- ☐ Project crashing
- ☐ Team management



PERT

- Activity durations are often uncertain.
- PERT incorporates variabilities in activity durations into task network analysis.
- Activity uncertainties are considered by using three time estimates for each activity.

Probabilistic Time Estimates

- PERT uses 3 time estimates for each activity:
- **Most likely time (m)**
 - subjective estimate of most frequent time
- **Optimistic time (a)**
 - shortest possible time (ideally)
- **Pessimistic time (b)**
 - longest time possible if everything went wrong

PERT Statistical Task Times

$$t_e = \frac{a+4m+b}{6} \quad s^2 = \frac{(b-a)^2}{36}$$

- a = optimistic time estimate
- m = most likely time estimate
- b = pessimistic time estimate ($a < m < b$)
- t_e = expected time for activity completion
- s^2 = variance of activity duration

Project Scheduling Tools

- OpenProj
- dotProject
- **GanttProject**
- Microsoft Projects
- PHProjekt
- ConsultComm
- TaskJuggler
- Collabtive
- OpenGoo
- ProjectPier
- Redmine

Project Crashing: Reducing Project Duration

- Often project manager faces the demand from customers for early delivery.
- Project manager needs to examine the critical path.
- Reduce the durations of the critical tasks by deploying more resources for the tasks.

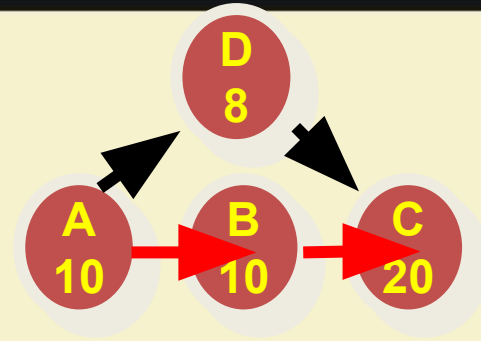
Project Crashing: A Simple Approach

1. Identify the critical path
2. Find cost per day to expedite each task on critical path.
3. For cheapest task to expedite, reduce it as much as possible, or until critical path changes.
4. Repeat 1-3 until no more possible.

Crashing Example

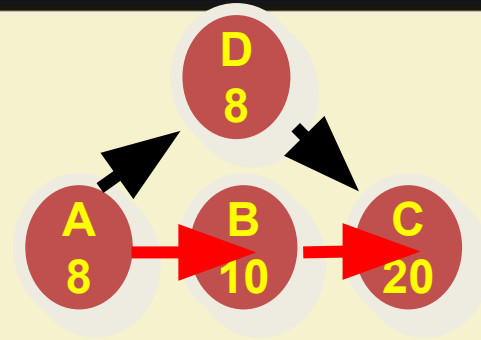
- ABC is critical path=40

	Crash cost per week avail		Crash wks
A	500	2	
B	1800	3	
C	2,000	2	
D	2,500	2	



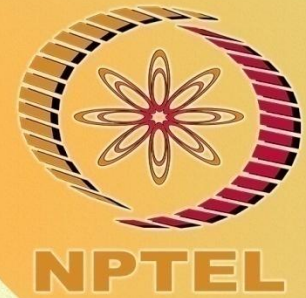
Crashing Example

- ABC is critical path=38



	Crash cost per week	Crash wks avail
A	500	0
B	1800	3
C	2,000	2
D	2,500	2

NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

Faculty Name: RAJIB MALL

Department : Computer Science and Engineering

TOPIC: Team Management



Managing a Project Team

- Tracking team member performance,
- Motivating team members,
- Providing timely feedback,
- Resolving issues and conflicts, and coordinating to help enhance the overall project performance.



Hawthorne effect

- 1920's – series of experiments at the Hawthorne Plant of Western Electric, Chicago
 - Worker productivity was measured at various levels of light illumination.
 - Researchers found that regardless of whether the light levels were raised or lowered, worker productivity increased.
 - **Concluded that simply showing an interest in a group increased productivity**



Selecting the best people

- Belbin distinguished between:
 - **Eligible** (having the right qualifications) and **suitable** candidates (can do the job).
- The danger is to employ someone who is eligible but not suitable.
- **The best situation:**
 - **Employ someone who is suitable but not eligible!**
 - **For example, they are likely to be cheaper and stay in the job.**

Organization Behaviour: Early Thought

- Frederick Taylor (1856-1915) 'the father of scientific management'
- Recommendation:
 - Select the best people for the job;
 - Instruct them in the best methods;
 - Give financial incentives in terms of pieces of work completed

McGregor's Theories

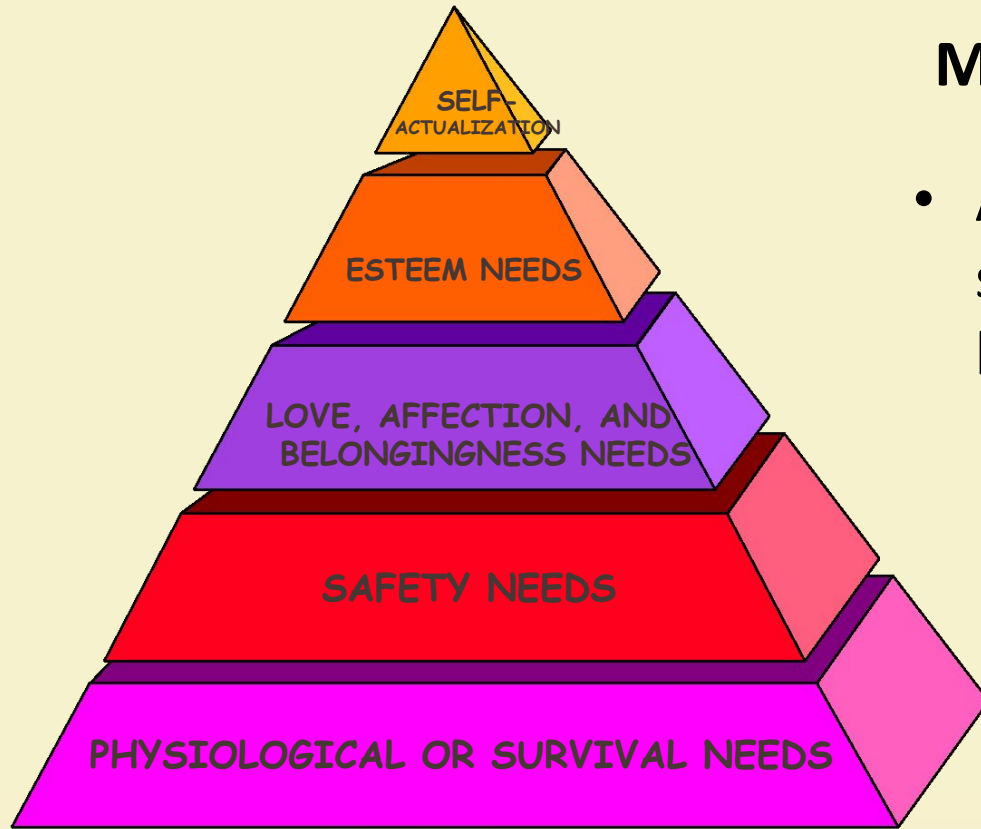
- **Theory X:** There is a need for coercion, direction, and control of people at work
- **Theory Y:** Work is as natural as rest or play

Do good software developers have innate characteristics?

- 1968 study – difference of 1:25 in time taken by different programmers to code a program
- Other research found experience better than maths skills as a guide to software skills
- Some research suggested software developers less sociable than other workers
- Later surveys have found no significant social differences between IT workers and others – this could be result of broader role of IT

Motivation

- Motivation and hard work can often make up for shortfalls in innate skills
- Taylor's approach - financial incentives
- **Abraham Maslow (1908-1970)**
 - Motivations vary from individual to individual
 - hierarchy of needs – as lower ones fulfilled, higher ones emerge
 - Lowest level – food, shelter
 - Highest level – self-actualization



Maslow's Hierarchy of Needs

- As a lower level of needs is satisfied --- gradually a higher level of need emerges.

Maslow's Hierarchy of Needs

- **Physiological needs:** Overcome hunger, thirst and sleep, etc.
- **Safety needs:** Security, protection from danger etc.
- **Social needs:** Friendship, engaging in social activities and group membership.
- **Esteem needs:** Self-confidence, achievement, recognition and appreciation.
- **Self-actualization:** The desire to develop and realize one's full potential.

Herzberg suggested that two sets of factors affected job satisfaction

Herzberg 2 Factor Theory

1. **Hygiene or maintenance factors** – make you dissatisfied if they are not right e.g. pay, working conditions
2. **Motivators** – make you feel the job is worthwhile e.g. a sense of achievement

- Motivating factors relate to content of job.
 - Called '**satisfiers**' and are factors which can increase the motivation to work better and harder.

Motivating factors

1. **Recognition of effort and performance**
2. **The nature of the job itself – does it provide the employee with an appropriate degree of challenge?**
3. **Sense of achievement**
4. **Opportunity for promotion**

Vroom

Vroom identified three influences on motivation:

1. ***Expectancy*** – The belief that working harder leads to better performance
2. ***Instrumentality*** – The belief that better performance will be rewarded
3. ***Perceived value of the reward***

- Identified the following characteristics of a job which make it more 'meaningful'

Oldham-Hackman job characteristics

- **Skill variety**
 - **Task identity**
 - **Task significance**
- Two other factors contributed to satisfaction:
 - **Autonomy**
 - **Feedback**

Methods to improve job satisfaction

- Set specific goals
- Provide feedback on the progress towards meeting those goals
- Consider job redesign
 - **Job enlargement**
 - **Job enrichment**

Stress

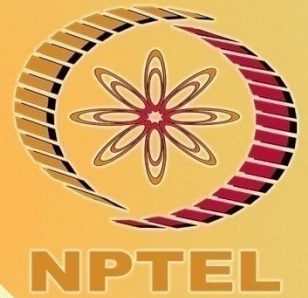
- Edward Yourdon quotes a project manager: *'Once a project gets rolling, you should be expecting members to be putting in at least 60 hours a week....The project manager must expect to put in as many hours as possible.'*
- 1960 study in US: people under 45 who worked more than 48 hours a week had twice the risk of death from coronary heart disease.
- **XP practice – maximum 40 hour working week**

- Good project management should lead to:
- Reasonable estimates of effort
- In turn to Good project control --- leading fewer unexpected crises
- Making clear what is expected of each team member – reduces **role ambiguity**
- Reduced **role conflict** where a person is torn between conflicting responsibilities
- **Bullying tactics are a symptom of incompetent project management.**

**Stress can be
reduced by good
project
management**

CONCEPTS COVERED

□ Organization and Team Structures

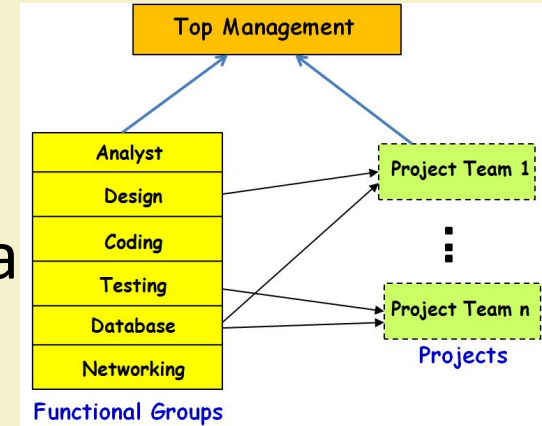


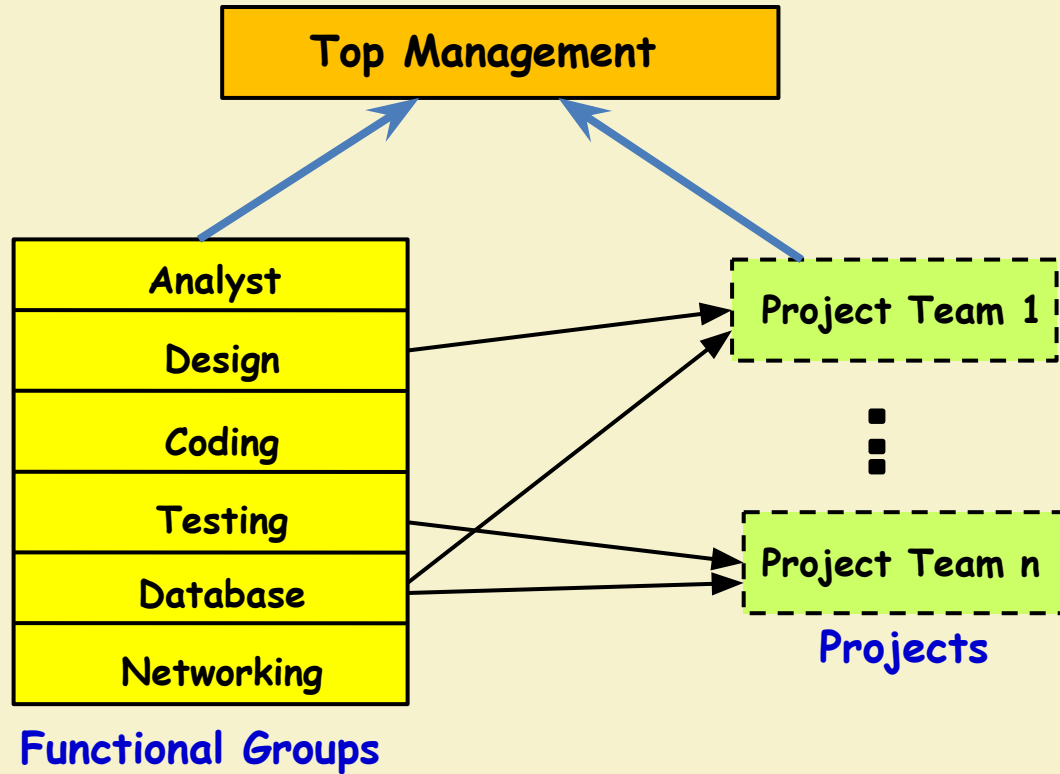
Organization and Team structures

- **Organization structure:**
 - How an organization is structured into teams
- **Team structure:**
 - How individuals in a team are structured
- **Typical organization structures are:**
 - **Functional organization**
 - **Project organization**
 - **Matrix organization**

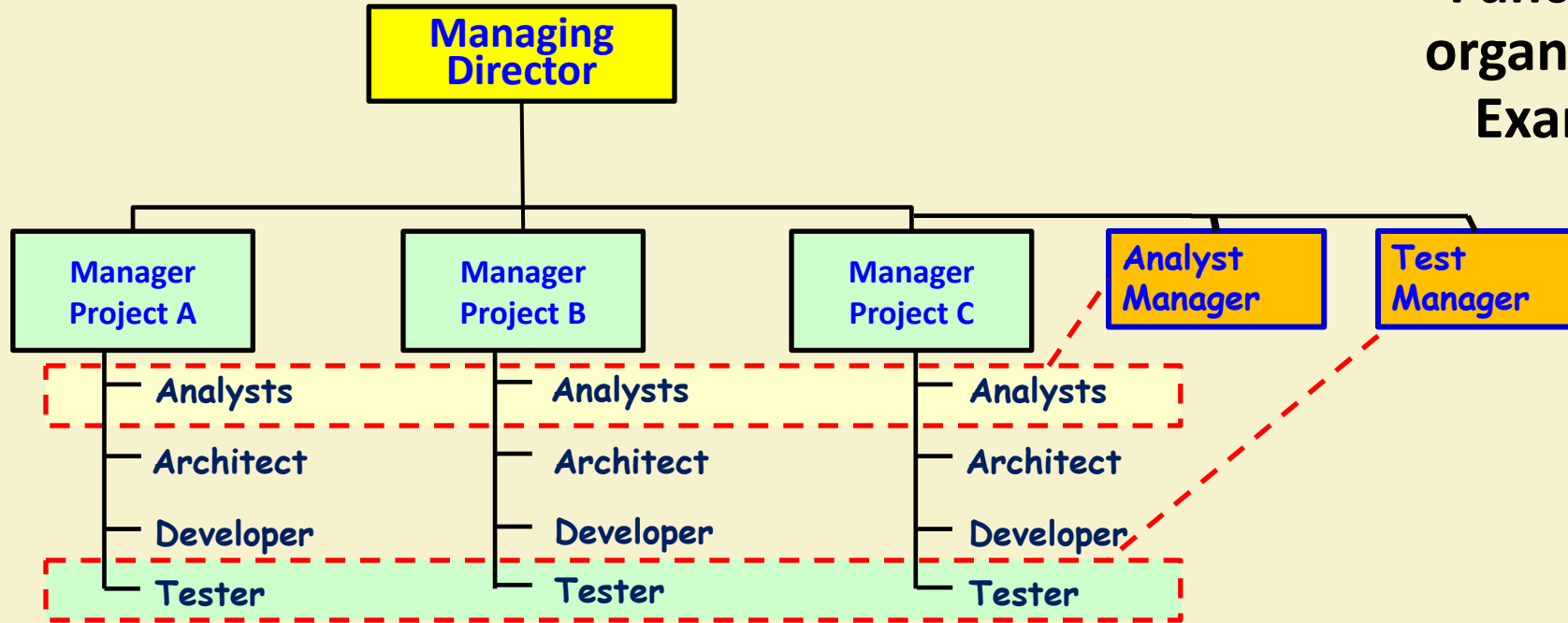
Functional organization

- Well known structure: Military, Police, hospitals,...
- **Within each functional group.**
 - Typically a pyramidal structure.
- Enables development of a better technical capability for the organization
 - Specialist groups share knowledge
- Projects become cost effective





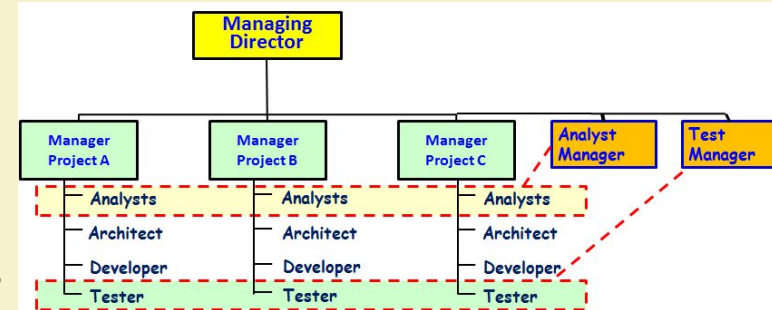
Functional organization Example



Advantages of functional organization

- **Specialists in the same area:**

- Better professional environment
- Cooperation between specialists in the same area



- **Maximum flexibility in the use of staff.**

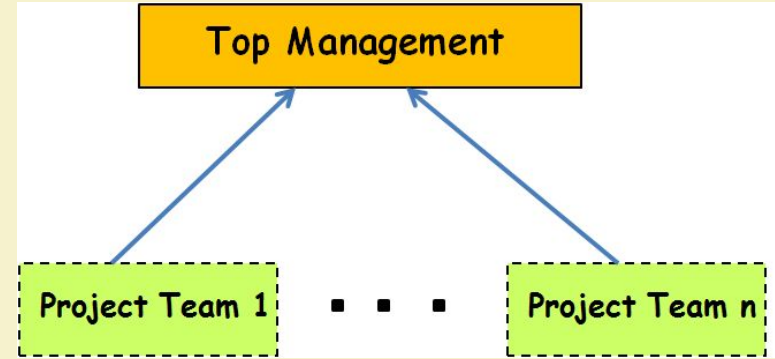
- Better manpower utilization.

Disadvantages of Functional Organization

- **Conflicts:**
 - Project manager and functional manager give different directives.
- **Project manager's control decreases**
- **Lack of job variety**

Project organization

- Each project has its own team.
- The project manager has full control on all team members.
- Team members stay on the project for its entire duration.



Advantages of Project Organization.

- Job rotation.
- Easy communication.
- Easier staffing.

Disadvantages of Project Organization

- Difficult to share expertise across projects...
- **Inefficient staffing**

Matrix organization

- A functional structure:
 - Over this exists a project structure.
- Two reporting (functional and project)
- Members report simultaneously to both functional and project managers.
- **Allows for member participation in multiple projects.**

Functional group	Project			
	#1	#2	#3	
#1	2	0	3	Functional manager 1
#2	0	5	3	Functional manager 2
#3	0	4	2	Functional manager 3
#4	1	4	0	Functional manager 4
#5	0	4	6	Functional manager 5
	Project manager 1	Project manager 2	Project manager 3	

Weak and Strong Matrix

- **Weak Matrix:**

- Project managers have complete control over the project budget
- Can accept or reject workers from functional group

- **Strong Matrix:**

- Functional managers assign workers and project managers would have to accept them

Matrix organization

- **Disadvantages:**

- Individuals have multiple bosses
- Project managers have reduced control
- Frequent shifting of workers in firefighting mode

- **Advantages:**

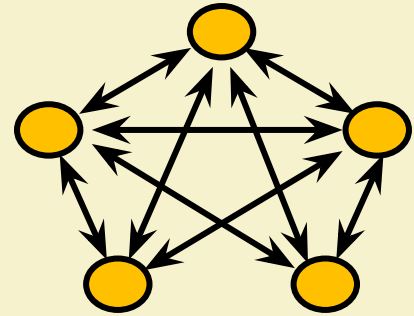
- Specialists belong to the same department and can consult each other easily.
- Better professional growth path.

Team Structure

- Three team structures are popular:
 - Democratic team
 - Chief programmer team.
 - Mixed organization

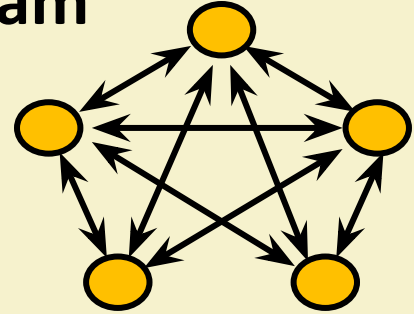
Democratic Team

- Typically small teams:
 - Decisions are based on consensus
 - Technical leader by consensus
 - Manager provides administrative leadership



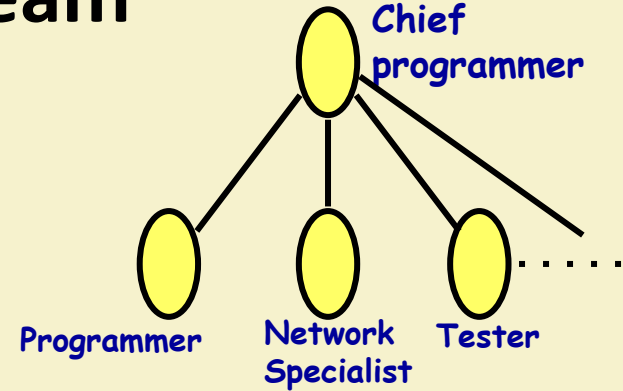
Egoless programming in a Democratic team

- Democratic teams are ideally egoless
- Ten or fewer programmers
- Ownership of code shared
- Exchange their code with other team members for review.



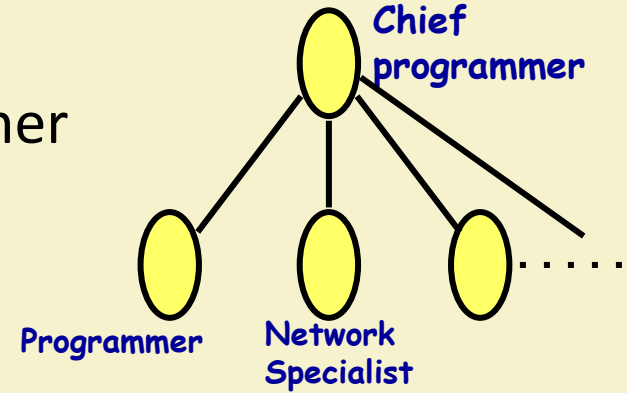
Chief programmer team

- Suitable for small teams.
- Chief programmer:
 - Manages all technical aspects.
 - Works out problem solutions.
 - Assigns tasks to the team members.
 - Reviews and monitors their work



Chief Programmer

- Both a manager *and* highly skilled programmer
- Carries out architectural design
- Allocates coding among the team members
- Writes the critical (or complex) sections of the code
- Handles all the interfacing issues
- Reviews the work of the team members
- Personally responsible for every line of code

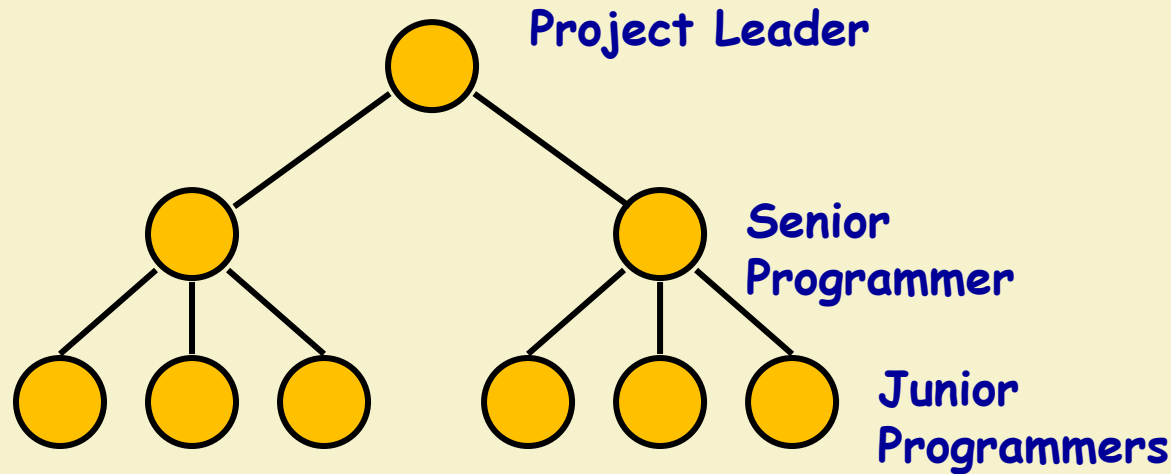


Large Projects

- For large projects:
 - **Neither chief programmer, nor democratic team is suitable.**
- Hierarchical team structure often used:
 - Junior programmers report to a senior programmer.
 - senior programmers report to a project leader.

Hierarchical Team

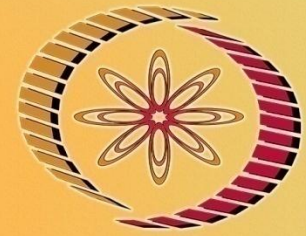
- Technical leadership is shared by the project leader and the seniors programmers.



Comparison of Team Structures

- **Chief programmer team:**
 - Best for low difficulty projects.
- **Democratic team:**
 - Communication overhead in a democratic team makes it inefficient as the team size increases.
- **Hierarchical team:**
 - Suitable for large projects.

NPTEL ONLINE CERTIFICATION COURSES



NPTEL

Course Name: Software Project Management

Faculty Name: RAJIB MALL

Department : Computer Science and Engineering

TOPIC: Risk Management



Few definitions of risk

'the chance of adverse consequences of future events'

PRINCE2

'an uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives'

PM-BOK

- Risks relate to **possible future** problems, not current ones
- They involve a possible event and its effect(s) e.g. developer leaves --> task delayed

A Broad Category of Risk

- Market risk
- Financial risk
- Technology risk
- Project risk: People, Schedule, etc

Risk Management Approaches

- **Proactive:**

- Anticipate the possible risks
- Take actions to manage the risks.

- **Reactive:**

- Take no action until an unfavourable event occurs.

A framework for dealing with risks

- **Risk identification** – what risks might there be?
- **Risk analysis and prioritization** – which are the most serious risks?
- **Risk planning** – what are we going to do about them?
- **Risk monitoring** – what is the current state of the risk?

Risk identification

- **Use of checklists** – usually based on the experience of past projects
- **Brainstorming** – getting knowledgeable stakeholders together to pool concerns
- **Causal mapping** – identifying possible chains of cause and effect

Boehm's top 10 development risks

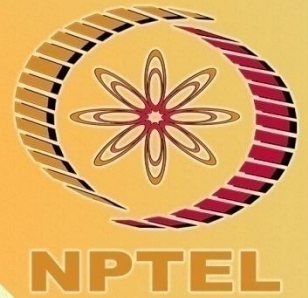
Risk	Risk reduction techniques
Personnel shortfalls	Staffing with top talent; training and career development;
Unrealistic time and cost estimates	Multiple estimation techniques; incremental development;
Developing wrong software functions	Improved software evaluation; user surveys; prototyping;
Developing wrong user interface	Prototyping; task analysis; user involvement

Boehm's top 10 risks

Gold plating	Requirements scrubbing, prototyping,
Late changes to requirements	Change control, incremental development
Shortfalls in externally supplied components	Benchmarking, inspections, formal specifications, contractual agreements, quality controls
Shortfalls in externally performed tasks	Quality assurance procedures, competitive design etc
Real time performance problems	Simulation, prototyping, tuning
Development technically too difficult	Technical analysis, cost-benefit analysis, prototyping , training

CONCEPTS COVERED

- ☐ Risk Management
- ☐ Software Quality Management



A framework for proactive risk management

- **Risk identification** – what risks might there be?
- **Risk analysis and prioritization** – which are the most serious risks?
- **Risk planning** – what are we going to do about them?
- **Risk monitoring** – what is the current state of the risk?

Risk prioritization

Risk exposure (RE)

= (potential damage) x (probability of occurrence)

Potential damage: a money value e.g. a flood would cause
Rs. 1 Billion of damage

Probability 0.00 (absolutely no chance) to 1.00 (absolutely
certain) e.g. 0.01 (one in hundred chance)

RE = 1 Billion x 0.01 = Rs. 10 million

Crudely analogous to the amount needed for an insurance
premium

Risk probability: qualitative descriptors

Probability level	Range
High	Greater than 50% chance of happening
Significant	30-50% chance of happening
Moderate	10-29% chance of happening
Low	Less than 10% chance of happening

Risk planning

Risks can be dealt with by:

- Risk acceptance
- Risk avoidance
- Risk reduction
- Risk transfer
- Risk mitigation/contingency measures

Risk reduction leverage

Risk reduction leverage =

$$(RE_{\text{before}} - RE_{\text{after}}) / (\text{cost of risk reduction})$$

RE_{before} is risk exposure before risk reduction e.g. 1% chance of a fire causing Rs. 200k damage

RE_{after} is risk exposure after risk reduction e.g. fire alarm costing Rs. 500 reduces probability of fire damage to 0.5%

$$RRL = (1\% \text{ of Rs}200\text{k}) - (0.5\% \text{ of Rs}200\text{k}) / \text{Rs}500 = 2$$

$RRL > 1.00$ therefore worth doing

Thank you

